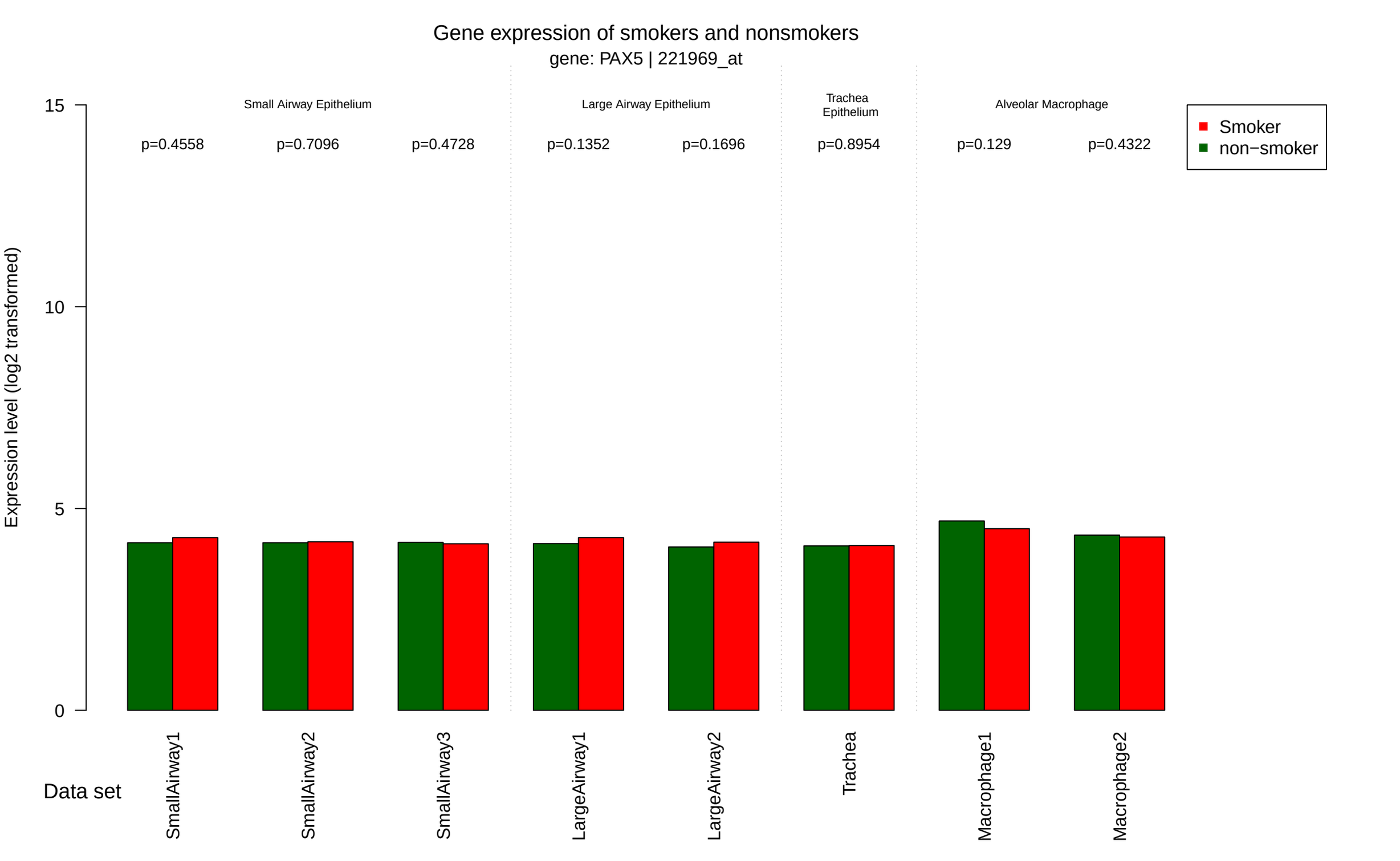
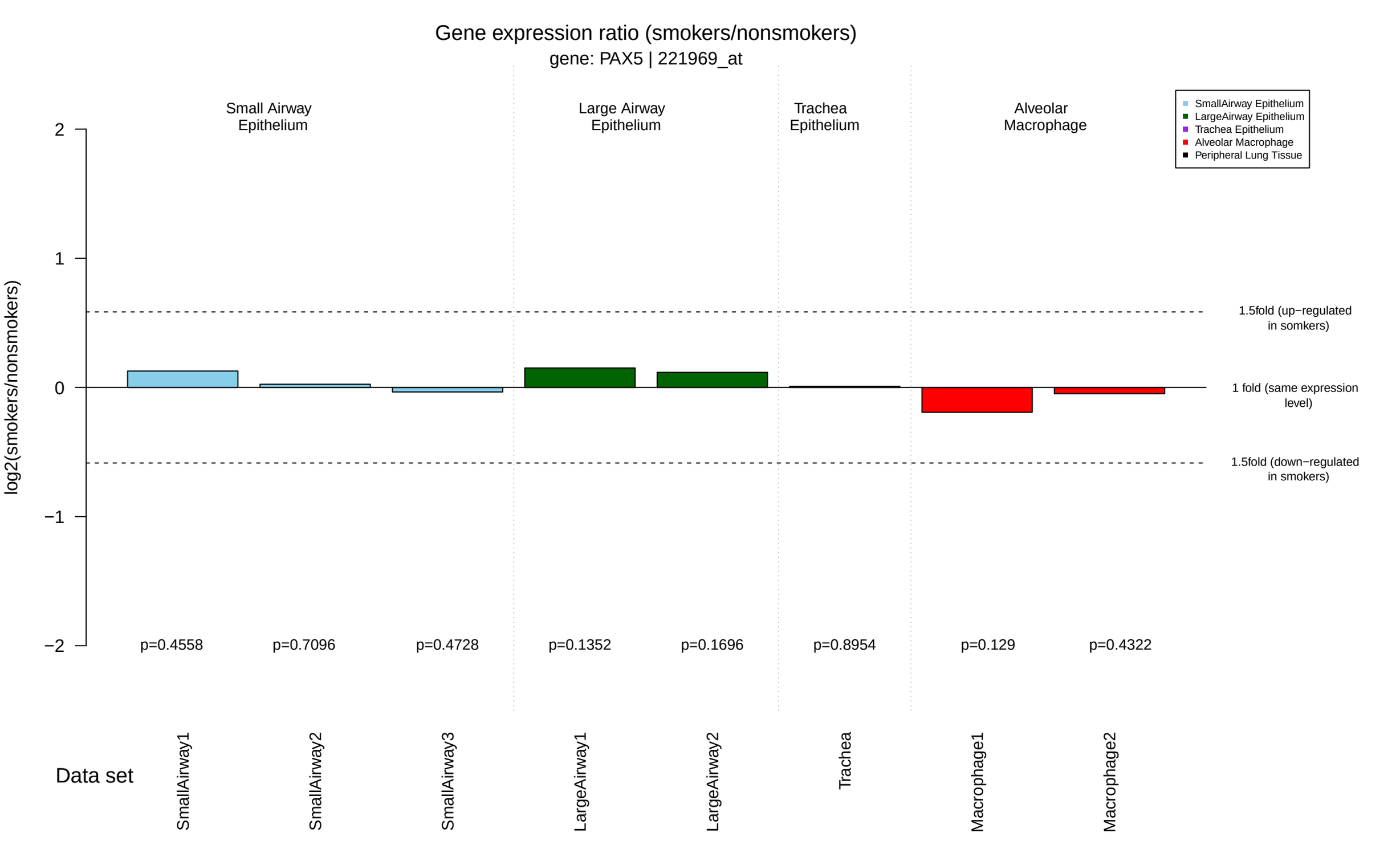
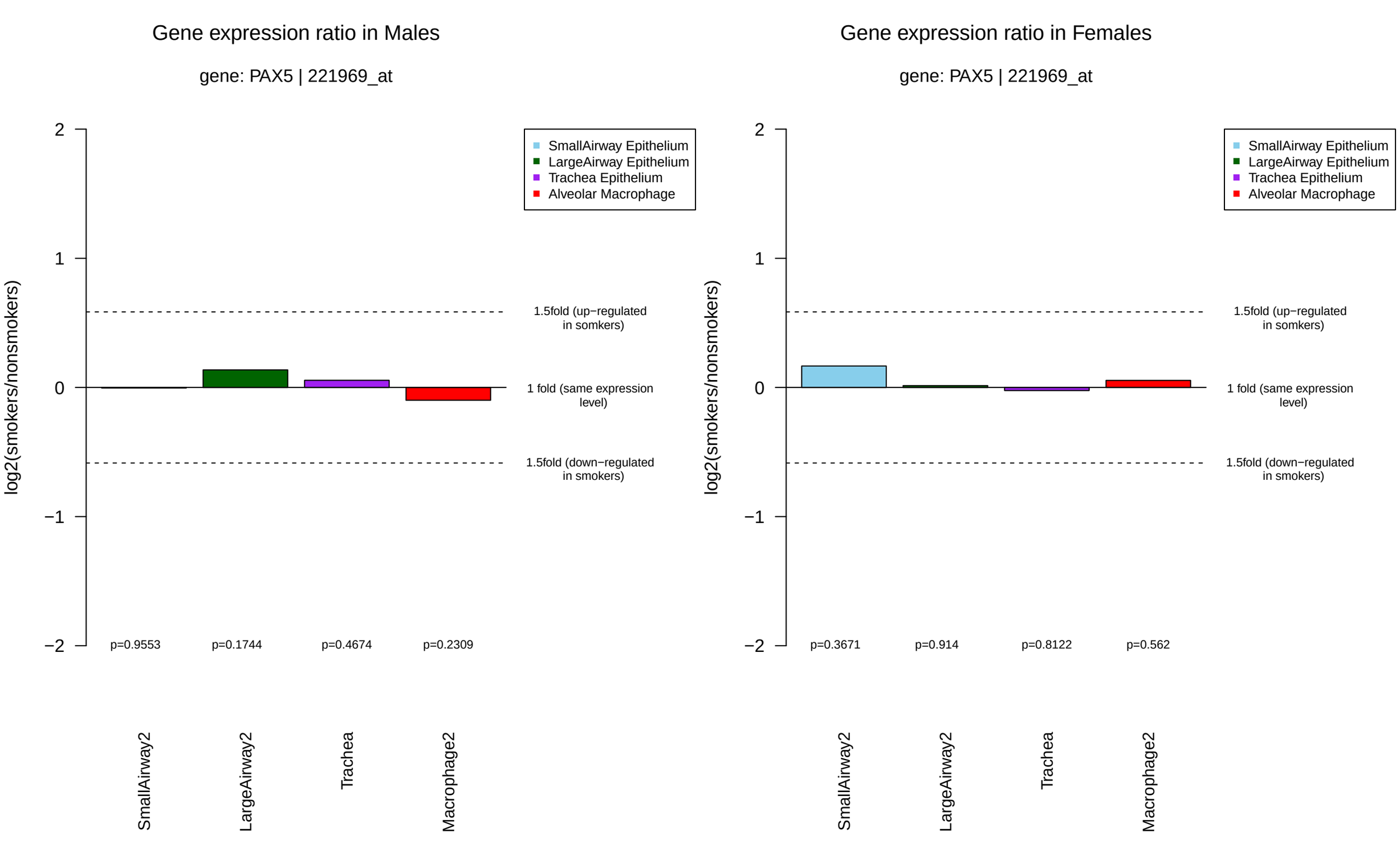
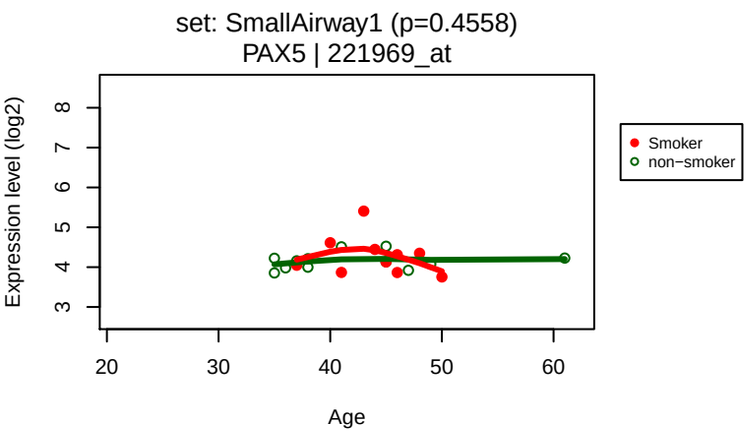
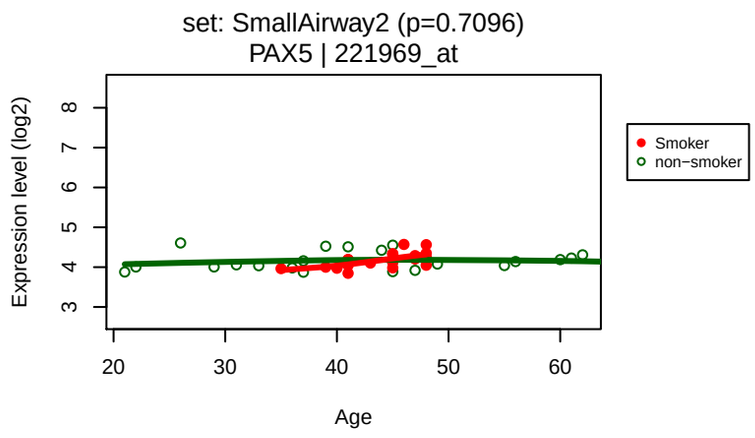
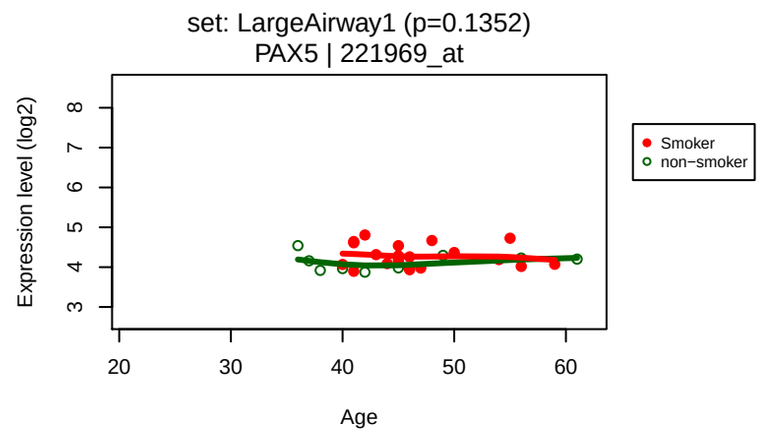

Gene expression of smokers and nonsmokers
gene: PAX5 | 221969_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4558
p=0.7096
p=0.4728
p=0.1352
p=0.1696
p=0.8954
p=0.129
p=0.4322
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

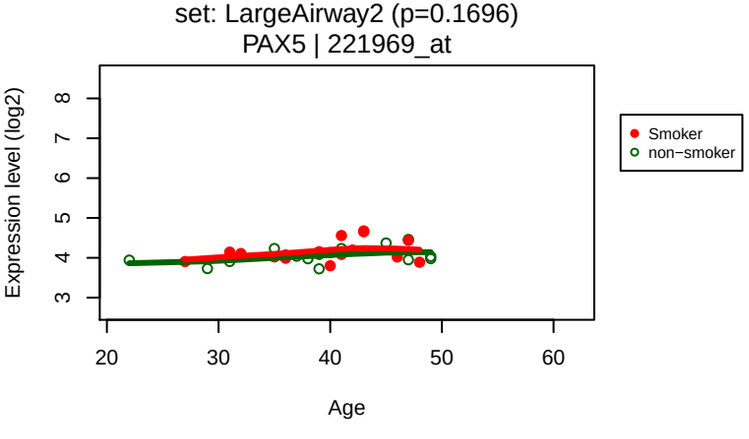

Gene expression ratio (smokers/nonsmokers)
gene: PAX5 | 221969_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4558
p=0.7096
p=0.4728
p=0.1352
p=0.1696
p=0.8954
p=0.129
p=0.4322
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

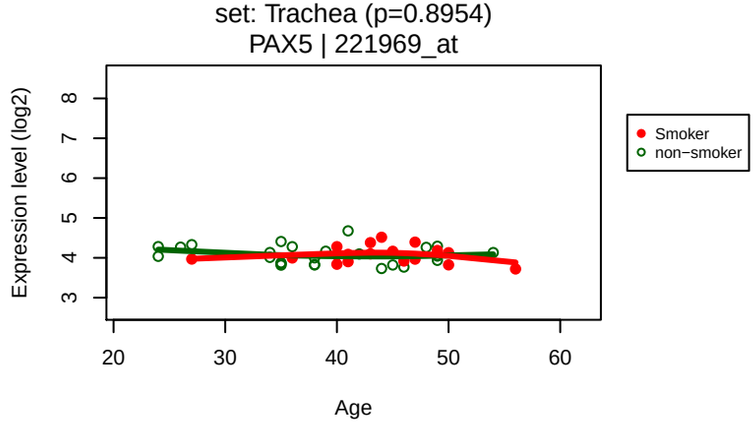

Gene expression ratio in Males
gene: PAX5 | 221969_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9553
p=0.1744
p=0.4674
p=0.2309
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAX5 | 221969_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3671
p=0.914
p=0.8122
p=0.562
SmallAirway2
LargeAirway2
Trachea
Macrophage2

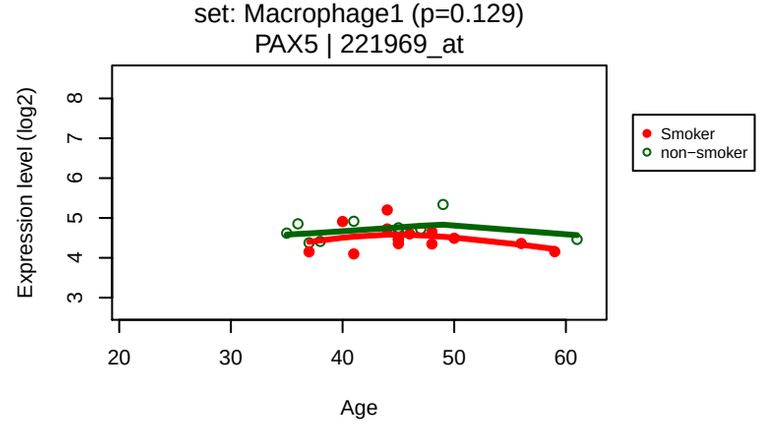

set: SmallAirway1 (p=0.4558)
PAX5 | 221969_at
Expression level (log2)
Age
Smoker
non−smoker

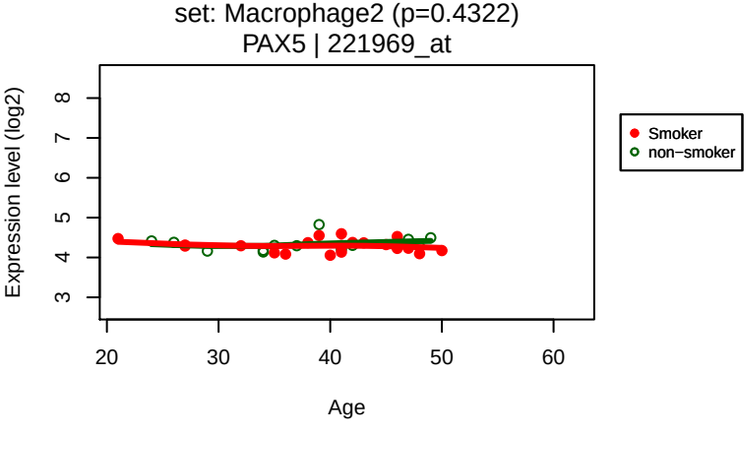

set: SmallAirway2 (p=0.7096)
PAX5 | 221969_at
Expression level (log2)
Age
Smoker
non−smoker

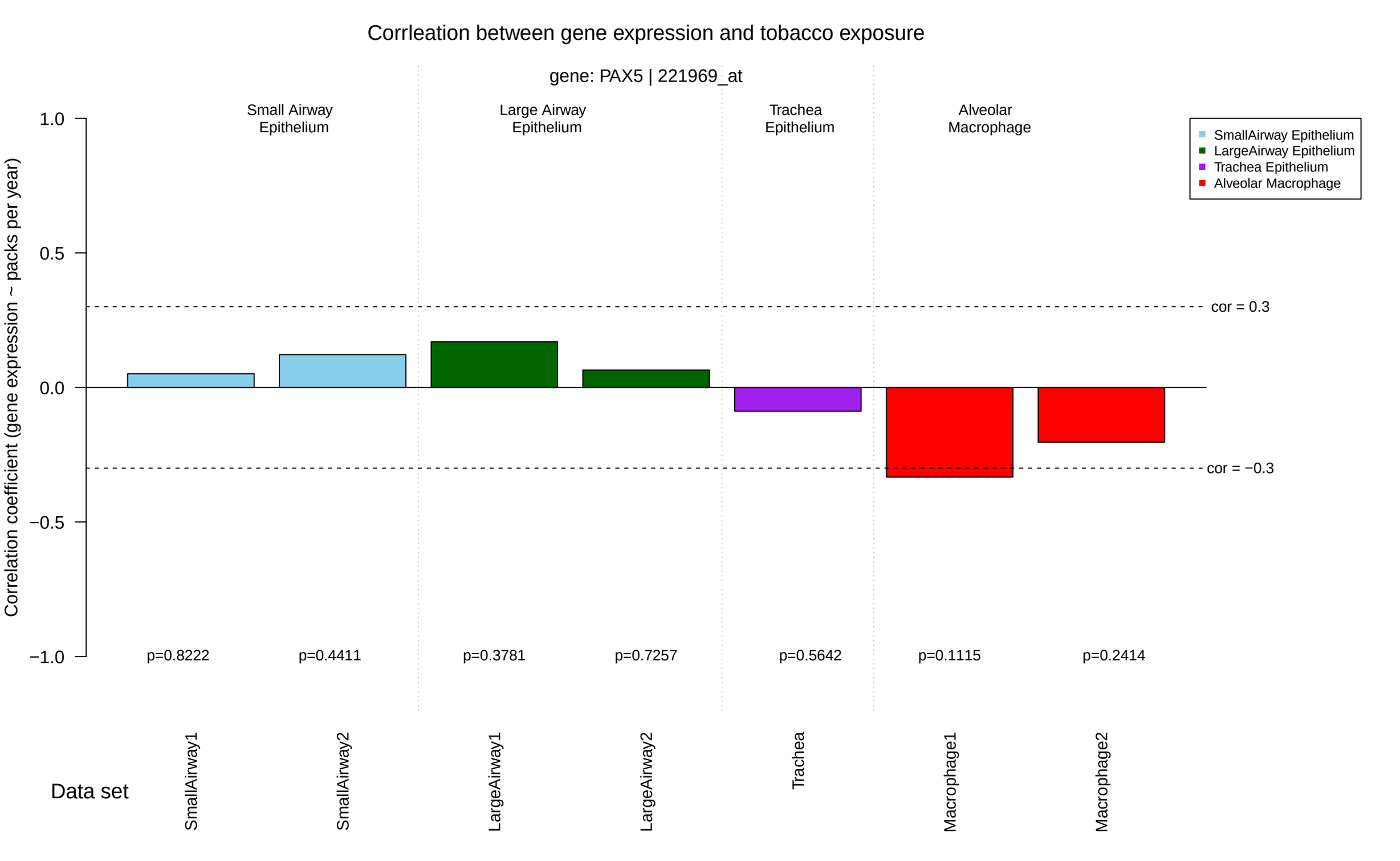

set: LargeAirway1 (p=0.1352)
PAX5 | 221969_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1696)
PAX5 | 221969_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8954)
PAX5 | 221969_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.129)
PAX5 | 221969_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4322)
PAX5 | 221969_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAX5 | 221969_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8222
p=0.4411
p=0.3781
p=0.7257
p=0.5642
p=0.1115
p=0.2414
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2